

Name: ID:

Pre-lab Questions: Measurements and Density Determination

1. Answer the following questions:

a. How many significant figures are in the following values?

i. 2804

ii. 5.029

iii. 0.003068

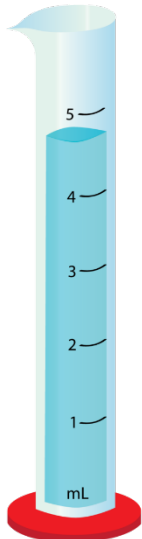
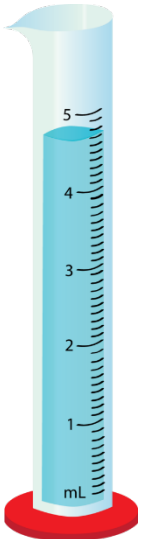
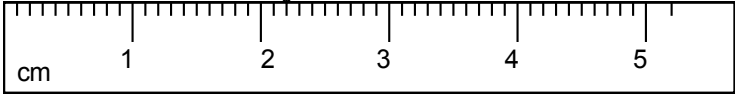
iv. 4.06×10^{-5}

v. 580.

vi. 75000

vii. 62.00

b. Record the reading of the following measuring devices to the correct number of significant figures.

Instrument		Reading
A  B 		A
		B
		

2. An empty graduated cylinder weighs 58.26 g. When a 5.00 mL sample of cleaning solution is added to the container, it weighs 65.41 g.
- Calculate the density of the cleaning solution to the correct number of significant figures. Show all your work.

- If the accepted density of the cleaning solution is 0.98 g/mL, calculate the percent (%) error.

$$\text{Percent error} = \frac{\text{Experimental value} - \text{Accepted value}}{\text{Accepted value}} \times 100\%$$

3. Answer the following questions about safety rules:
- List two items of P.P.E. (Personal Protective Equipment) that must be worn **at all times** in the lab.
 - What two things should you do if a chemical comes in contact with your eye?
 - If a piece of glassware is accidentally broken, where should it be thrown away?